EETS 7304

Module 4 Part 1
UDP & TCP

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SMU

Topics/Objectives

- User Datagram Protocol (UDP)
- Transport Control Protocol (TCP)

User Datagram Protocol

UDP uses, characteristics, parameters and applications

UDP/TCP

- Connection-oriented versus connectionless
- UDP is connectionless
  - Ideal for real-time applications
    - Voice over IP
    - Video
    - IPTV
    - Streaming media
    - Trivial File Transfer Protocol
    - Online gaming
    - Domain name System (DNS)
    - Dynamic Host Configuration Protocol (DHCP)
    - Multicast
- TCP is connection-oriented

UDP Parameters

- UDP was defined in RFC 768 and refined in RFC 1122.
- All implementations must follow both RFCs to make interoperability reliable.
- UDP uses IP protocol ID 17.
- Any IPv4 or IPv6 packet received with 17 in the protocol ID field is given to the local UDP service.

UDP Parameters

- UDP is defined as stateless (no session information is kept by hosts) and unreliable (no guarantees of any QoS parameters, not even delivery).
- This does *not* mean that UDP traffic is somehow lower priority on the network or through routers.
- It’s not as if UDP traffic is routinely tossed by stressed-out routers; it just means that if the application using UDP needs to keep track of a session history if it is going to happen.

UDP Parameters

1 byte	1 byte	1 byte	1 byte
Source Port		Destination Port	
Length (including header)		Checksum	
Datagram Data (optional)			

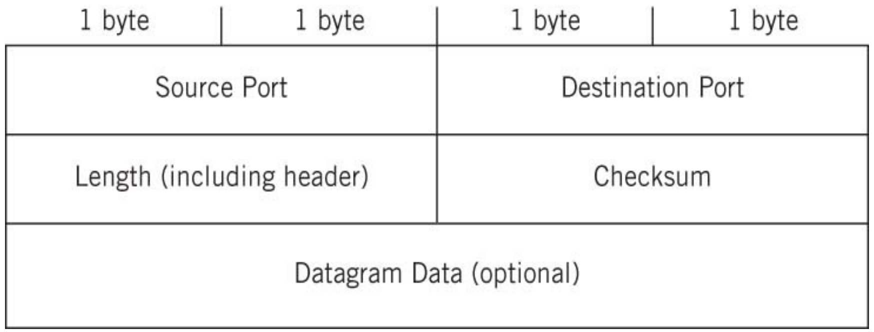
UDP Header: Port Numbers

- The header is only 8 bytes (64 bits) long.
- First are the 2-byte Source Port field and the 2-byte Destination Port field.
  - These fields are the datagram counterparts of the source and destination IP addresses at the packet level.
  - But unlike IP addresses, there is no structure to the port fields: All values between 0 and 65,353 are represented as pure numeric.
  - This does not mean that all port numbers, source and destination, are the same, however.
  - Port values can be divided into well-known, registered, and dynamic port numbers.

1 byte	1 byte	1 byte	1 byte
Source Port		Destination Port	
Length (including header)		Checksum	
Datagram Data (optional)			

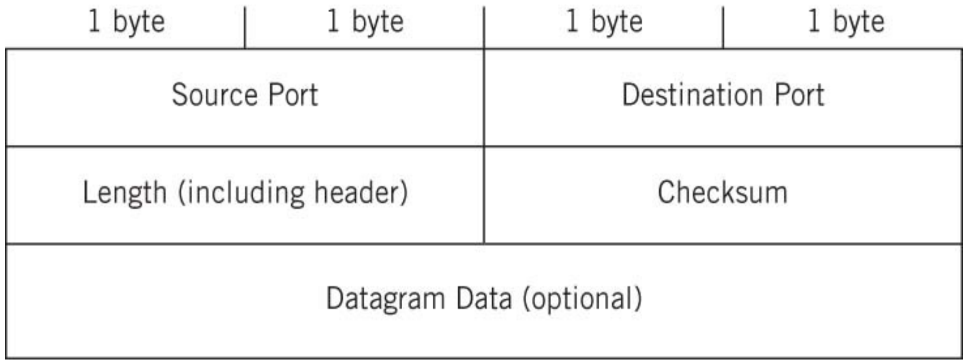
UDP Header: Length

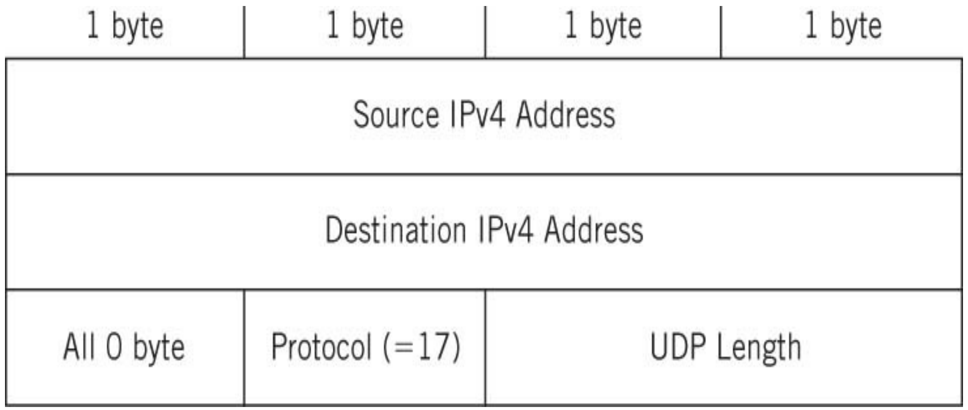
- The Length field gives the length in bytes of the UDP datagram, and includes the header fields along with any data.
- The minimum length is 8 (the header alone), and the maximum value is 65,353.
- However, the achievable maximum UDP datagram lengths are determined by the size of the send and receive buffers on the host end systems, which are usually set to around 8000 bytes (although they can be changed).



UDP Header: Checksum

- The Checksum field is the most interesting field in the UDP header.
- This is because the checksum is *not* a simple value calculated on the UDP header fields and data, if present.
- The UDP checksum is computed on what is called the *pseudo-header*.
- Includes fields in the IP header.





UDP Header: Checksum IPv6

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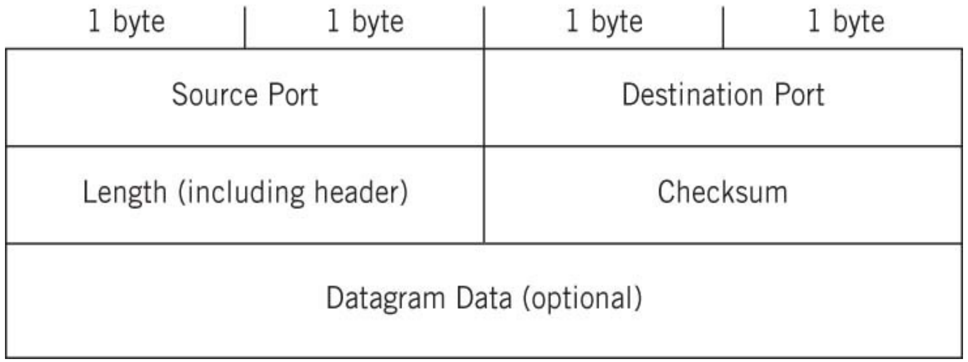
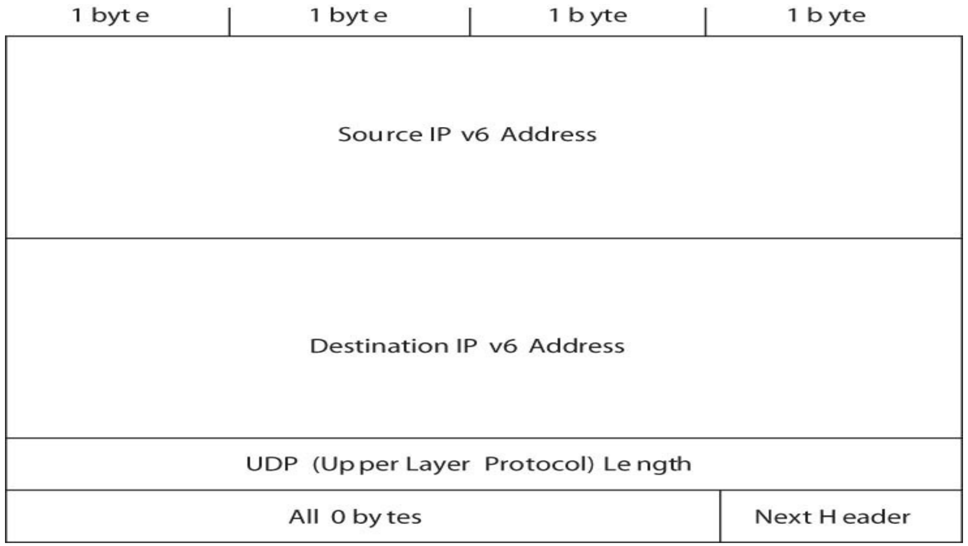
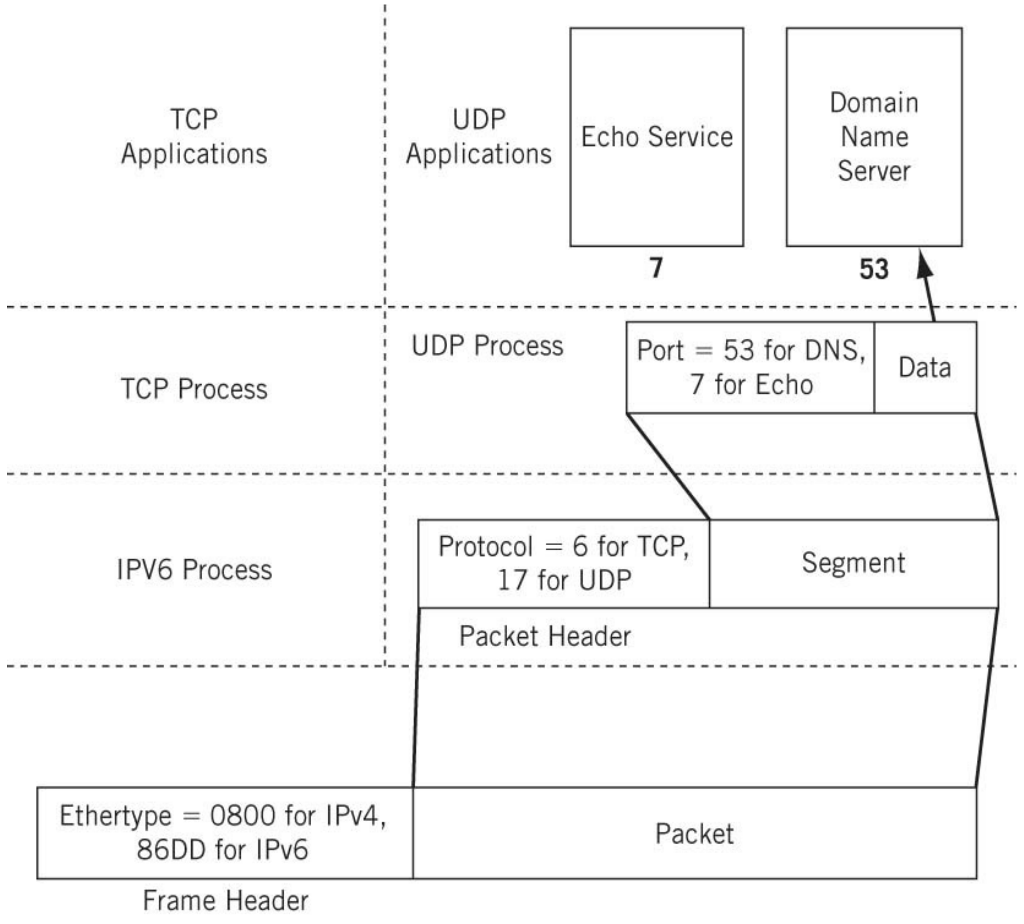
- The UDP checksum is computed on what is called the *pseudo-header*.
- Includes fields in the IP header.
- Optional in IPv4, mandatory in IPv6.





Port Numbers

- Each application running above UDP (and TCP) and IP is indexed by its port number, allowing for the multiplexing of the IP layer.



Port Numbers

- *Well-known ports*—Ports in the range 0 to 1023 are assigned and controlled.
- *Registered ports*—Ports in the range 1024 to 49151 are not assigned or controlled, but can be registered to prevent duplication.
- *Dynamic ports*—Ports in the range 49152 to 65535 are not assigned, controlled, or registered. They are used for temporary or private ports.

Port Numbers

Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

Registered TCP Ports:

1863 MSN Messenger
8008 Alternate HTTP
8080 Alternate HTTP

Well Known TCP Ports

21 FTP
23 Telnet
25 SMTP
80 HTTP
110 POP3
194 Internet Relay Chat (IRC)
443 Secure HTTP (HTTPS)

Sockets

- The combination of IPv4 or IPv6 address and port numbers forms an abstract concept called a *socket*.
- For example, 10.10.12.166:80.

Transport Control Protocol (TCP)

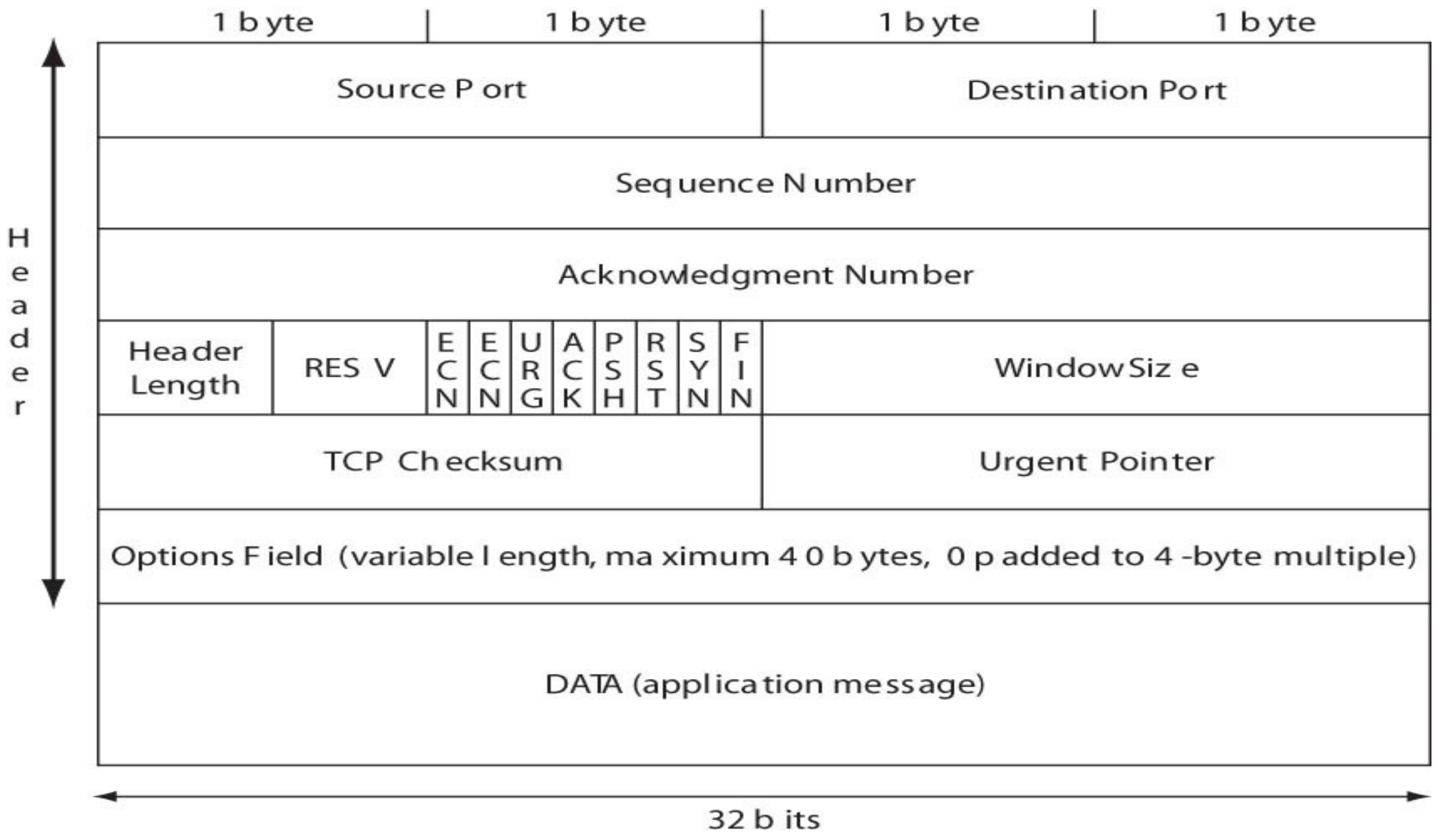
TCP Characteristics, fields and operations

Transmission Control Protocol (TCP)

- The Transmission Control Protocol (TCP) is as complex as UDP is simple.
- Some of the same concepts apply to both because both TCP and UDP are end-to-end protocols.
- Sockets and ports, well-known, dynamic, and private, apply to both.
- The major difference between UDP and TCP is that TCP is connection oriented.

Transmission Control Protocol (TCP) Header

- The TCP header is the same for IPv4 and IPv6



TCP Fields

- *Source and destination port*—49,152 through 65,535 is more in line with current standards.
- *Sequence number*—Each new connection (re-tries of failed connections do not count) uses a different *initial sequence number* (ISN) as the basis for tracking segments.
- *Acknowledgment* Number-of next segment receiver expects.
- *Header length*—The TCP header length in 4-byte units.
- *Reserved*—Four bits are reserved for future use.

TCP Fields

- *ECN flags*—The two explicit congestion notification (ECN) bits are used to tell the host when the network is experiencing congestion and send windows should be adjusted.
- *URG, ACK, PSH, RST, SYN, FIN*—These six single-bit fields (Urgent, Acknowledgment, Push, Reset, Sync, and Final) give the receiver more information on how to process the TCP segment.

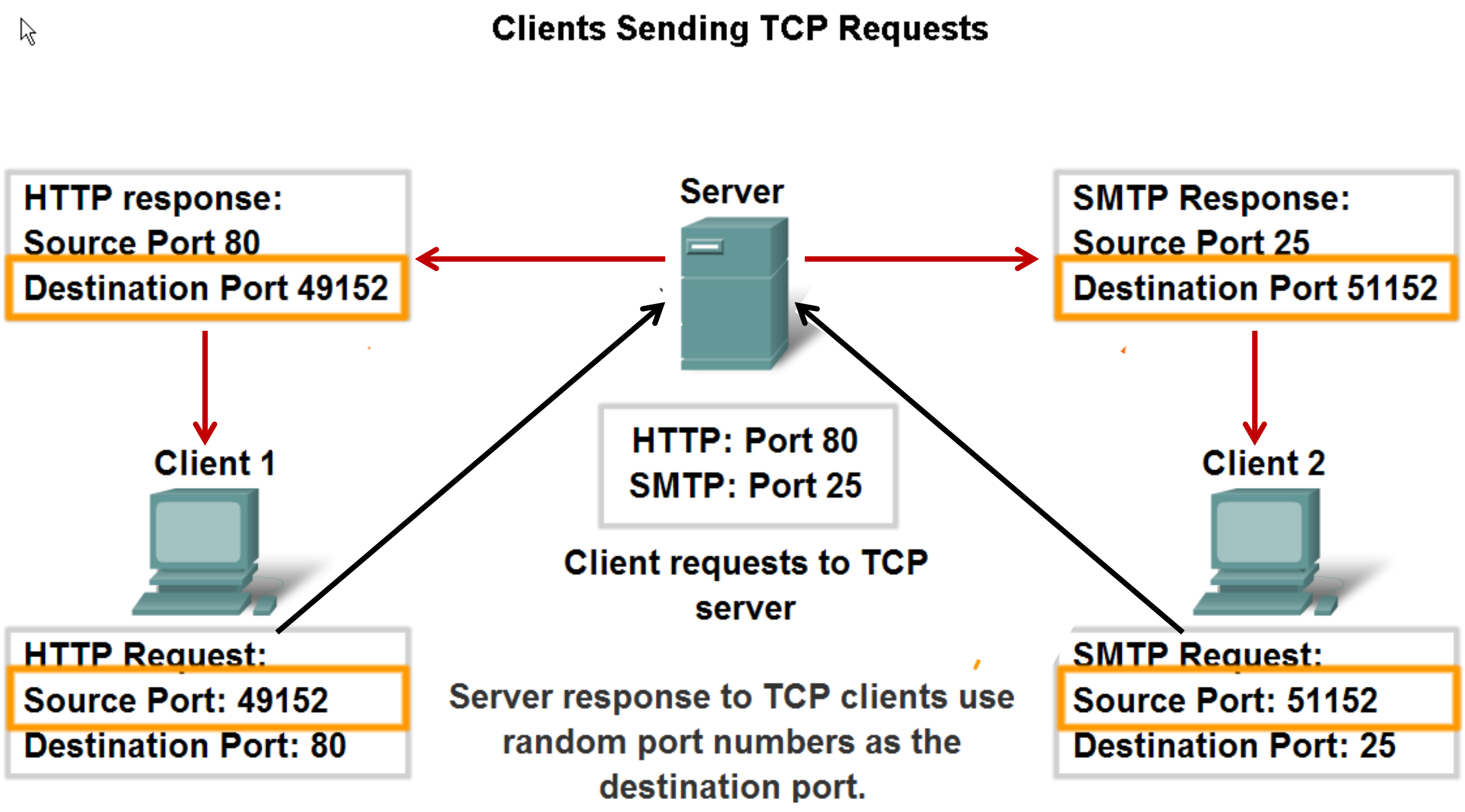
TCP Fields

- *Window size*—The size of receive window that the destination host has set.
  - This field is used in TCP flow control and congestion control. It should not be set to zero in an initial SYN segment.
- *Checksum*—An error-checking field on the entire TCP segment and header as well as some fields from the IP datagram (the pseudo-header).
  - The fields are the same as in UDP. If the checksum computed does not match the received value, the segment is silently discarded.

TCP Fields

- *Urgent pointer*—If the URG control bit is set, the start of the TCP segment contains important data that the source has placed before the "normal" contents of the segment data field.
- *Options and padding*—TCP options are padded to a 4-byte boundary and can be a maximum of 40 bytes long.

TCP Operation: Ports



TCP Characteristics

- TCP is a virtual circuit service that adds reliability to the IP layer, reliability that is lacking in UDP.
- TCP also provides sequencing and flow control to the host-to-host interaction, which in turn provides a congestion control mechanism to the routing network as a whole (as long as TCP, normally an end-to-end concern, is aware of the congested condition).
- The flow control mechanism in TCP is a ***sliding window*** procedure that prevents senders from overwhelming receivers and applies in both directions of a TCP connection.

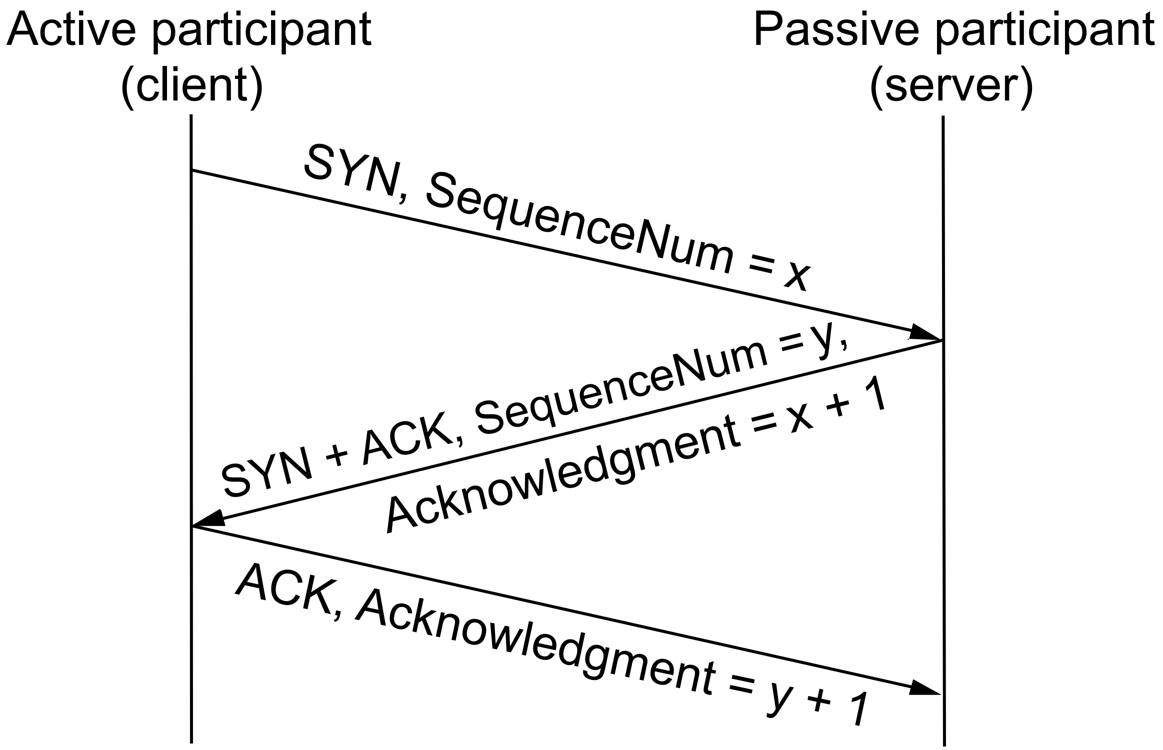
TCP Operation: Data Exchange

- TCP establishes end-to-end connections over the unreliable, best-effort IP packet service using a special sequence of three TCP segments sent from client to server and back called a *three-way handshake*.
- Once the three segments are exchanged, data transfer can take place from host to host in either direction.
- Connections can be dropped by either host with a simple exchange of segments.

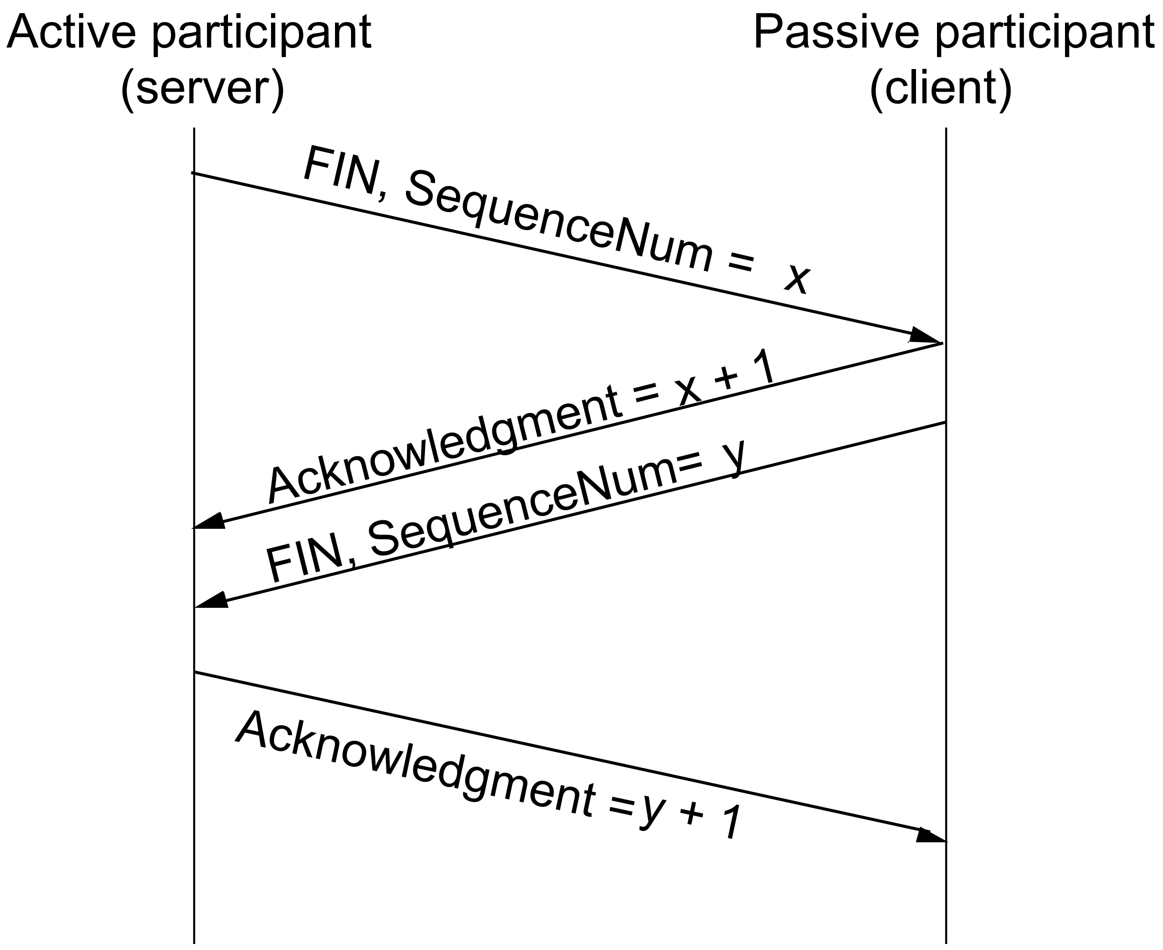
TCP Operation: Data Exchange

- TCP uses unique terminology for the connection process. A single bit called the SYN (synchronization) bit is used to indicate a connection request.
- This single bit is still embedded in a complete 20-byte (usually) TCP header, and other information, such as the *initial sequence number* (ISN) used to track segments, is sent to the other host.
- Connections and data segments are acknowledged with the ACK bit, and a request to terminate a connection is made with the FIN (final) bit.

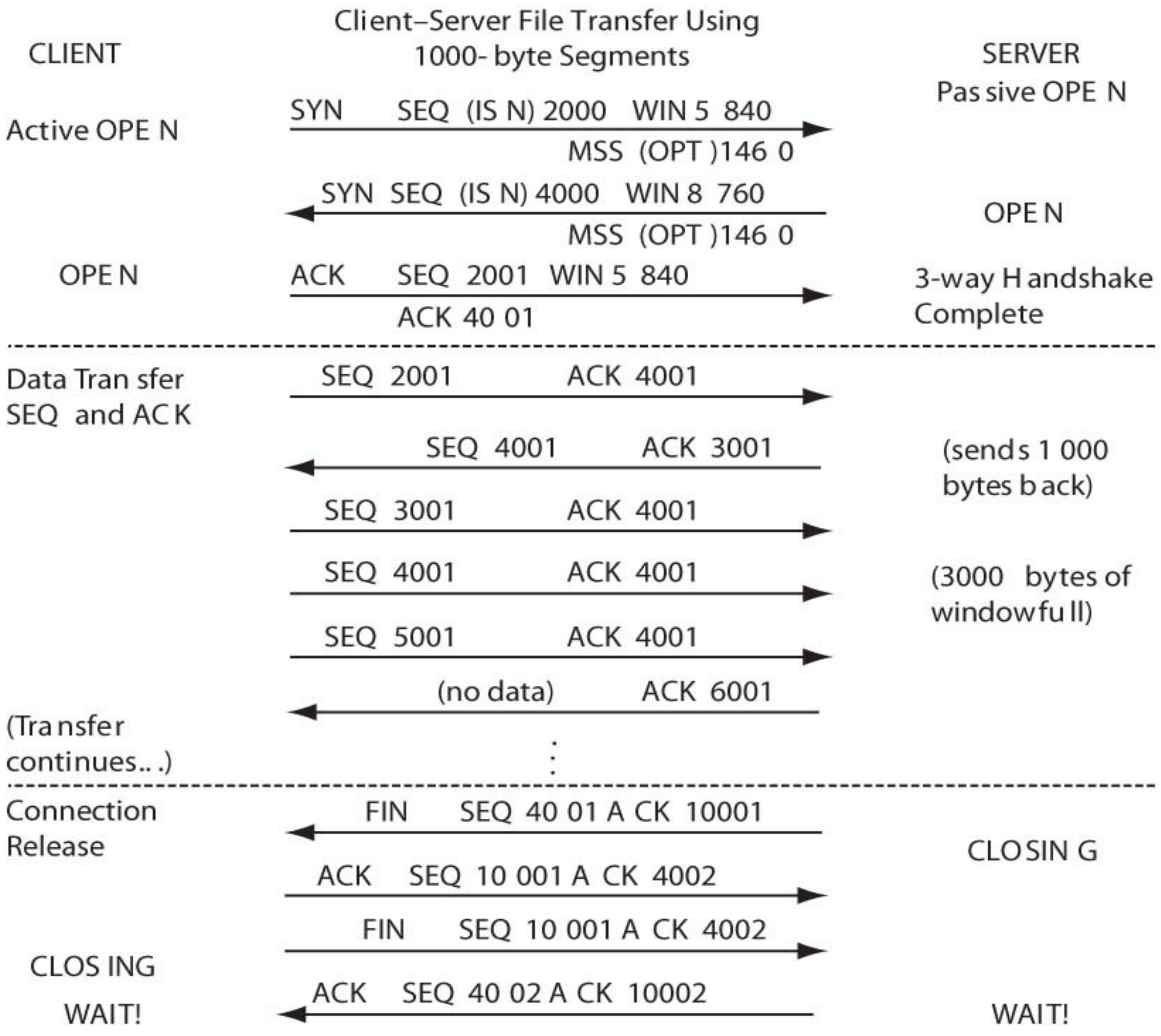
Connection Establishment



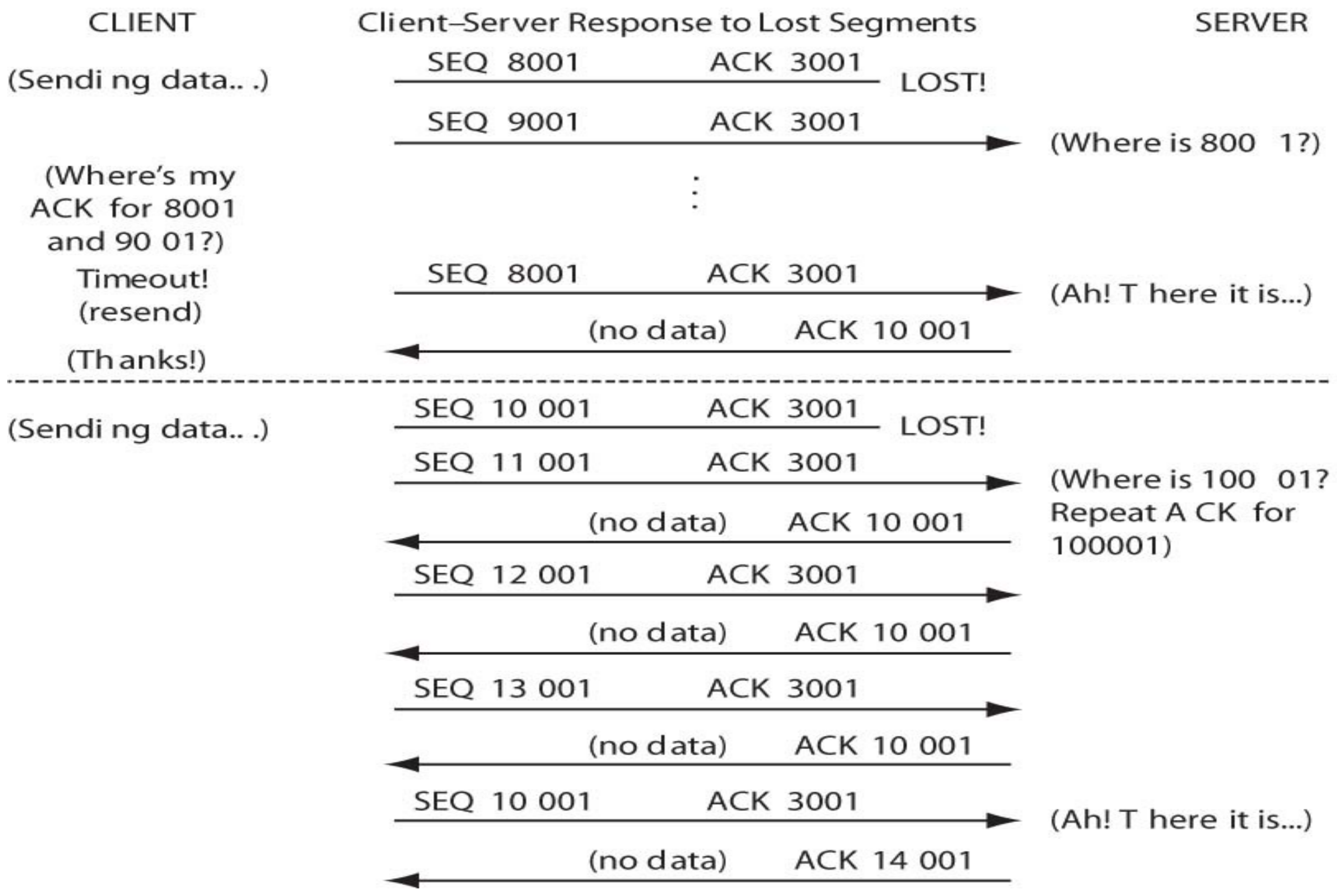
Connection Termination



TCP Operation: Data Exchange



TCP Operation: Data Operation



TCP Operation: Flow Control

- Sliding window – used by TCP to control how much data the sender will send before getting an acknowledgment, and works to avoid traffic congestion by the receiver letting the sender know to reduce the size of data it is sending between acknowledgments during periods of congestion or high latency.

See UDP/TCP Ports on Host

- Netstat –a

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	HOSTNAME:0	LISTENING
TCP	192.168.1.190:56963	52.230.222.68:https	ESTABLISHED
TCP	192.168.1.190:57086	HPFC3FDB20B07C:8080	ESTABLISHED

Summary/Takeaways

- Explain the characteristics of UDP, when it is used, and why it is used.
- Capture, diagram and analyze a TCP traffic flow that shows the 3-way handshake, and its use of sequence number and acknowledgements.
- Complete Internet Protocol (IP) Lab#6.